

MAX BEAN-TIERNEY

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EDUCATION

Brown University

Expected: May 2028

Bachelor of Science in Mechanical Engineering

- GPA 3.9/4.0 Relevant coursework: Electrical Circuits and Signals, Statics and Dynamics, Thermodynamics, Control Systems

EXPERIENCE

Tripathi Laboratory, Brown University

May 2025 – Present

Lead Engineer, Gantry & Control Systems – Vivvity (developed in collaboration with Revvity)

- Led mechanical design and systems integration of gantry and electromagnetic locking mechanism for Vivvity, an automated microfluidic liquid-handling instrument developed with Revvity as the successor to the LabChip GxII Touch
- Designed and validated electromagnetic clamping system; electromagnets on either side of the microfluidic chip interfacing with a 1018 steel plate mounted to electrode block; confirmed reliable seal up to 30 psi with Ansys and mechanical test rig
- Engineered dual lead-screw gantry system across horizontal and vertical axes, constraining vertical height to avoid interference with main liquid-handler pipette toolhead; designed custom hardware to reduce system footprint
- Developed embedded firmware (Marlin/C++) and Python GUI, enabling closed-loop pressure control, live telemetry, and crash prevention logic

Brown Space Engineering

Jan 2025 – Present

Engineer

- Designed and tested mechanical systems for deployment on a student-developed CubeSat, set to be launched in May

Shelfelbine Laboratory, Northeastern University

May 2023 – August 2023

Research Assistant

- Developed a MATLAB program to automate analysis of millions of joint reaction force data points
- Reduced processing time from weeks to hours for a biomechanics study on knee osteoarthritis treatments
- Performed statistical analysis and presented findings to faculty and collaborators, directly informing treatment outcomes

PROJECTS

Automated Pressurization Module for Reproducible Microfluidic Fragment Analysis

Technologies: Solidworks, additive manufacturing, low-throughput microfluidics

- Developed a chip priming device using a closed-loop pressure sensing system to improve reproducibility of assays
- Designed a novel gasket sealing around the sides of the well rather than against its rim, achieving 5x greater contact area for a more reliable, consistent seal
- Presented results at the Society for Laboratory Automation and Screening (SLAS) International Conference and Exhibition

Patent-Pending Electroosmotic Flow Cell Device — Co-Inventor

Technologies: electrokinetics, rapid prototyping

- Designed and 3D-printed a proof-of-concept flow cell using two glass coverslips and a custom 0.1mm PTFE gasket, forming a 0.1mm channel with integrated electrode access points
- Validated feasibility using a dyed gel-buffer solution under an applied electric field, demonstrating electroosmotic flow through the channel; results were presented to faculty and PhD researchers and directly informed the device's continued development into its current patent-pending form

TECHNICAL SKILLS

Software: Solidworks, Marlin, Ansys products, MATLAB, COMSOL, Python, Microsoft Office

Hardware: Additive manufacturing/3D printing, motor control/implementation, pneumatic devices, prototyping

Concepts: Mechanical design, design for assembly, design for manufacturing, industrial design, technical drawings